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Student Submissions — CO2_AT3_Coding Challenge

SIMATS Engineering • CSE • CSA0802 — Python Programming • 29-07-2026 • Category: CO2_AT3_Coding Challenge • Student Submissions Report

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Q1. String Character Frequency Counter

CODE

```
s = input().lower()

freq = {}

for ch in s:
    if ch != " ":
        if ch in freq:
            freq[ch] += 1
        else:
            freq[ch] = 1

for ch in sorted(freq):
    print(f"{ch} → {freq[ch]}")
```

INPUT

Hello World

OUTPUT



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Q2. Vowel and Consonant Classifier with Statistics

CODE

```
s = input().lower()

vowels = "aeiou"
vowel_count = 0
consonant_count = 0

vf = {}
cf = {}

for ch in s:
    if ch.isalpha():
        if ch in vowels:
            vowel_count += 1
            vf[ch] = vf.get(ch, 0) + 1
        else:
            consonant_count += 1
            cf[ch] = cf.get(ch, 0) + 1

print("Vowels:", vowel_count)
print("Consonants:", consonant_count)

if vf:
    mv = max(vf, key=vf.get)
    print("Most frequent vowel:", mv, "(", vf[mv], ")")
else:
    print("Most frequent vowel: None")

if cf:
    lc = min(cf, key=cf.get)
    print("Least frequent consonant:", lc, "(", cf[lc], ")")
else:
    print("Least frequent consonant: None")

if consonant_count > 0:
    print("Vowel-to-Consonant Ratio:", round(vowel_count / consonant_count, 2))
else:
    print("Vowel-to-Consonant Ratio: Undefined")
```

INPUT

Python Programming

OUTPUT

```
Vowels: 4
Consonants: 13
Most frequent vowel: o ( 2 )
Least frequent consonant: y ( 1 )
Vowel-to-Consonant Ratio: 0.31
```

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Q3. String Compression and Run-Length Encoding

CODE

```
s = input()

if s == "":
    print("Compressed:")
    print("Decompressed:")
else:
    compressed = ""
    count = 1

    for i in range(1, len(s)):
        if s[i] == s[i - 1]:
            count += 1
        else:
            compressed += s[i - 1]
            if count > 1:
                compressed += str(count)
            count = 1

    compressed += s[-1]
    if count > 1:
        compressed += str(count)

    if len(compressed) < len(s):
        print("Compressed:", compressed)
    else:
        print("Compression not beneficial.")
        print("Original:", s)

    decompressed = ""
    i = 0
    while i < len(compressed):
        ch = compressed[i]
        i += 1
        num = ""
        while i < len(compressed) and compressed[i].isdigit():
            num += compressed[i]
            i += 1
        if num == "":
            decompressed += ch
        else:
            decompressed += ch * int(num)

    print("Decompressed:", decompressed)
```

INPUT

aabbccddddee

OUTPUT

Compressed: a2b3c2d4e2
Decompressed: aabbccddddee

Q4. Advanced Text Statistics Analyser

CODE

```

import re

text = input()

sentences = len(re.findall(r'[.!?]', text))
if sentences == 0:
    sentences = 1

paragraphs = len([p for p in text.split('\n') if p.strip()])

words = re.findall(r'\b[a-zA-Z]+\b', text)
word_count = len(words)

avg_word = round(sum(len(w) for w in words) / word_count, 2) if word_count else 0
avg_sentence = round(word_count / sentences, 2)

stop = {"the", "is", "a", "an", "in", "of", "and", "to", "for"}

freq = {}
for w in words:
    w = w.lower()
    if w not in stop:
        freq[w] = freq.get(w, 0) + 1

top = sorted(freq.items(), key=lambda x: (-x[1], x[0]))[:5]

vowels = "aeiou"

def syllables(word):
    count = 0
    prev = False
    for c in word.lower():
        if c in vowels:
            if not prev:
                count += 1
            prev = True
        else:
            prev = False
    return max(1, count)

total_syllables = sum(syllables(w) for w in words)

score = 206.835 - 1.015 * (word_count / sentences) - 84.6 * (total_syllables / word_count)

print("Words:", word_count)
print("Sentences:", sentences)
print("Paragraphs:", paragraphs)
print("Avg Word Length:", avg_word)
print("Avg Sentence Length:", avg_sentence)

print("Top 5 Words:", end=" ")
for w, c in top:
    print(f"{w}({c})", end=" ")
print()

print("Flesch Reading Ease:", round(score, 1))

```

INPUT

Python is a powerful programming language. It is easy to learn. It supports data science. Python is widely used.

OUTPUT

```

Words: 19
Sentences: 4
Paragraphs: 1
Avg Word Length: 4.74
Avg Sentence Length: 4.75
Top 5 Words: it(2) python(2) data(1) easy(1) language(1)
Flesch Reading Ease: 68.4

```